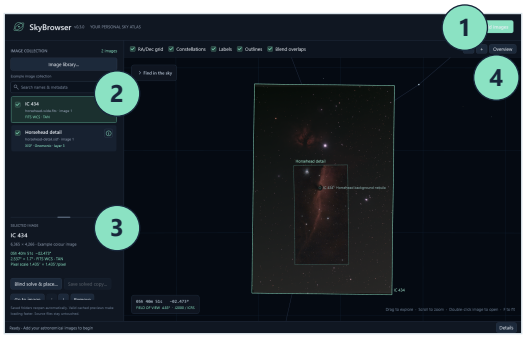
SkyBrowser

Beginner's user manual

Explore your astronomical photographs in their place on the sky. SkyBrowser combines your FITS and XISF images with a simple star map. You can look around, compare photographs and return to your saved image folders later.



The main window, showing a wide Horsehead field and a smaller image of the same region.

- **1. Top bar:** see the app version, open Options and add images.
- **2. Image collection:** open the library, search loaded images and choose one.
- **3. Selected image:** change the chosen image's appearance, visibility and placement. Drag the divider above this panel to resize it.
- **4. Sky view:** drag to look around and scroll to zoom. The bottom status bar reports progress and problems.

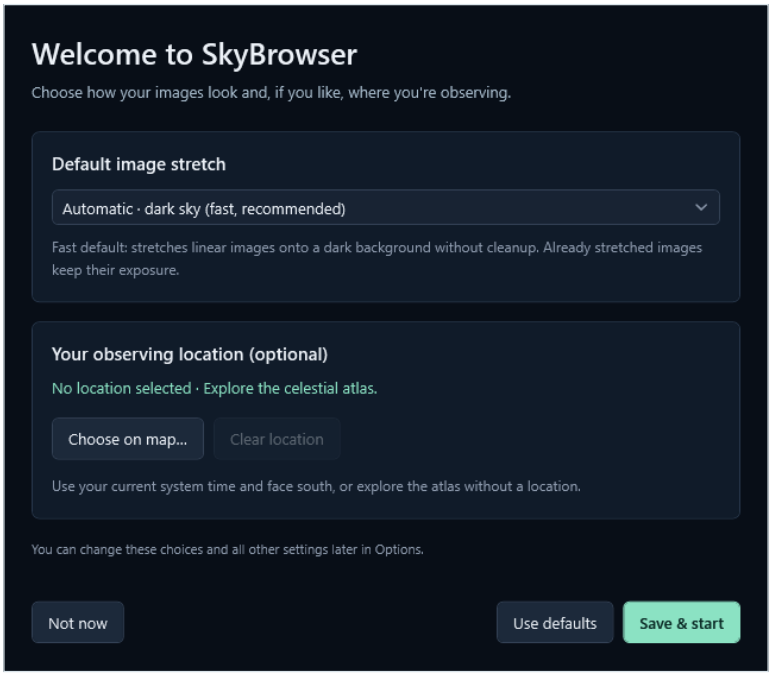
Start here	Page	Explore further	Page
Load your first images	2	Overlapping photographs	6
Navigate and find a target	3	Location and observing time	7
Collection and image details	4	Options and the cache	8
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Your originals stay safe. Display changes, blending and placement do not rewrite your scientific image files. Keep your originals available for future starts.

01 / Your first images

Start with the defaults

1. Open **SkyBrowser.exe** from the published application folder.
2. On the welcome screen, keep **Automatic · dark sky (fast)**. It makes dark, unstretched data easier to see without the slower cleanup steps.
3. Your observing location is optional. Leave it empty to explore the sky as an atlas, or choose **Choose on map...** to set it now. You can change it later in Options.
4. Choose **Save & start**. **Use defaults** also saves the factory choices. **Not now** continues without saving setup; it can appear again next time.



The welcome screen appears when no saved settings or recovery backup exists.

Load and frame a photograph

1. Choose **Image library**, then **Add folder...** to build your list of image folders.
2. Choose **Load folders** to load their images, including subfolders. Then double-click a collection entry, or use **Go to image**, to frame it.
3. Drag the sky to move around and scroll to zoom in or out.

Use **+ Add images** or drop files onto the window for individual imports. Your saved folders reopen automatically on startup, using cached previews when valid and loading source files when needed. See page 14 for folder management.

Supported files: .fits, .fit, .fts, .xisf and .xish. Keep distributed XISF headers with their data files. Images need a usable sky solution to appear automatically; see page 11 for **Needs placement** and ASTAP solving.

Recognized Bayer metadata automatically produces a color preview. Missing or unsupported masks stay grayscale; original files are unchanged.

02 / Explore the sky

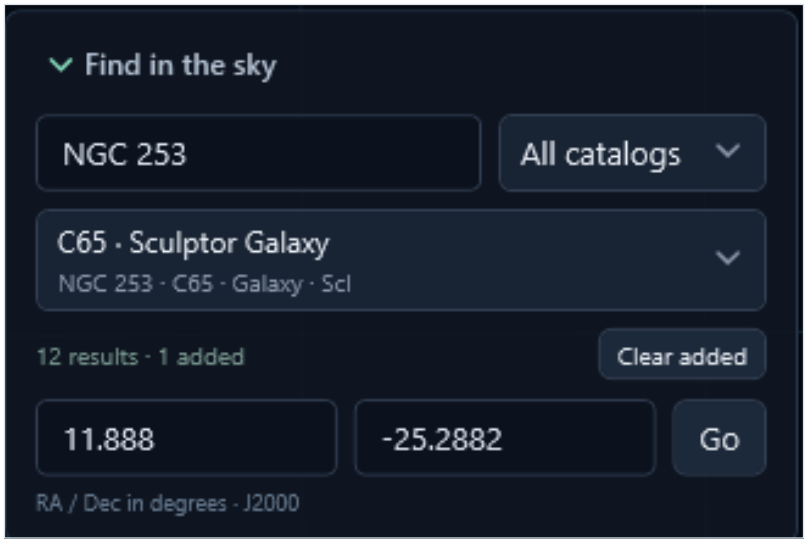
Move around

Action	How to do it
Look in another direction	Drag an empty part of the sky with the left mouse button.
Zoom	Scroll the mouse wheel, or use the + and - buttons above the sky.
Frame the selected image	Choose Go to image, or press F while the sky has focus.
Return to a wide view	Choose Overview, or press Home while the sky has focus.

The switches control the grid, constellation lines, labels and image outlines. Atlas mode uses **RA/Dec grid**; an observing location switches to **Alt/Az grid**, fixed to the local horizon.

Find a named object

1. Expand **Find in the sky** in the upper left of the sky view.
2. Type a name or catalogue number, such as **M31**, **IC 434**, **NGC 253** or **Horsehead**.
3. Optionally choose Messier, Caldwell, NGC or IC to narrow the catalogue.
4. Open the results list and choose an object. The sky turns toward it, adds a marker and fills in its coordinates.



The expanded finder. NGC 253 and Caldwell 65 are names for the same galaxy.

The finder includes Messier and Caldwell objects plus NGC and IC entries with known positions. Initially, only a small selection of prominent targets is labelled. **Clear added** removes the markers you have added; it keeps your photographs.

The shared **Go** button returns to the selected target. You can also enter a known **RA** and **Dec**, then choose Go. Both input boxes use decimal degrees. The sky's RA readout may use hours, so do not paste an hours/minutes/seconds value into the degree input. Collapse the finder again to recover space.

> **A catalogue marker is a signpost.** Finding an object does not download its photograph. Add your own image files to see photographs at that position.

03 / Your image collection

Choose, hide and search

Click an image's entry to show its controls under **Selected image**. Its checkbox hides or shows it on the sky. **Remove** takes it out of this collection; it does not delete the source file. The arrow buttons reorder entries. Drag the horizontal divider to give the collection or the selected-image controls more room.

Use **Search names & metadata** to find an image by filename, target, telescope, filter or other retained metadata. **ic434**, **IC 434** and **IC-434** match the same target. Multiple words narrow the results: try **IC 434 H-alpha**. Press **Ctrl+F** to focus search; the x button or Escape in the search box clears it.

> **Search filters the list.** An image hidden from the search results can still be visible on the sky. Use its checkbox to hide it from the sky itself.

Inspect the information behind a photograph

Choose the small circled **i** beside an entry to open **Image details**. The window shows its preview, source path, original and preview dimensions, file size, display mode and sky placement. Search the metadata table to find a keyword, value or comment. Select a row to read and copy its value in the box below.

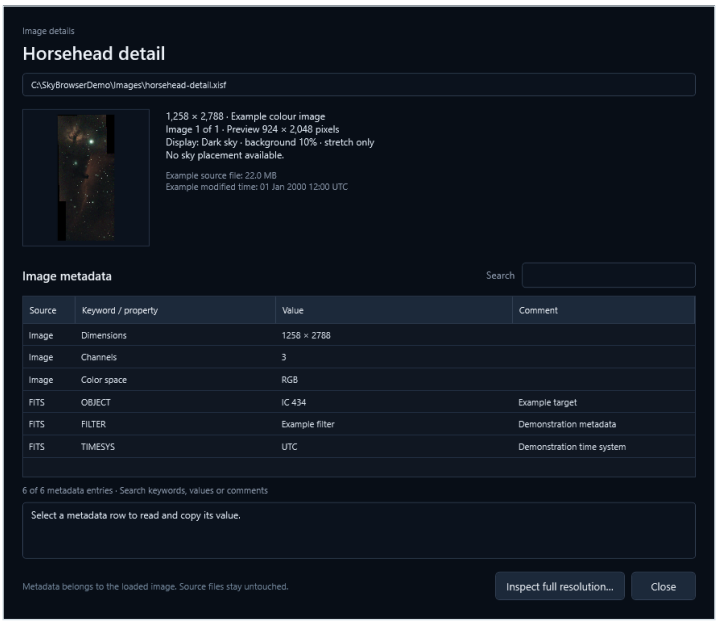
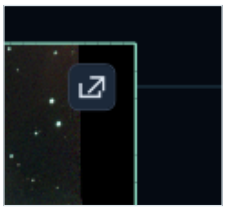


Image details for the narrower Horsehead image. Metadata and the original source path are available together.

Double-click a photograph in the sky to inspect it internally, choosing the foremost overlapping image. **Inspect full resolution…** here or under Selected image also opens 100% zoom, source values, sky coordinates, histograms and statistics (page 15).

To open the original in the Windows default application, use the small arrow button in a corner of the hovered or selected sky image. Zoom in if the photograph is too small to show it. Large metadata values may be shortened.



The corner arrow opens the original in its default application. Double-click the photograph itself for internal inspection.

04 / Make images easy to see

Many astronomical files store **linear**, or unstretched, data. They can look almost black until a display stretch makes faint detail visible. SkyBrowser changes the preview, while keeping your source file intact.

Select an image and use its **Display** list. Changing a mode may take a moment to process.

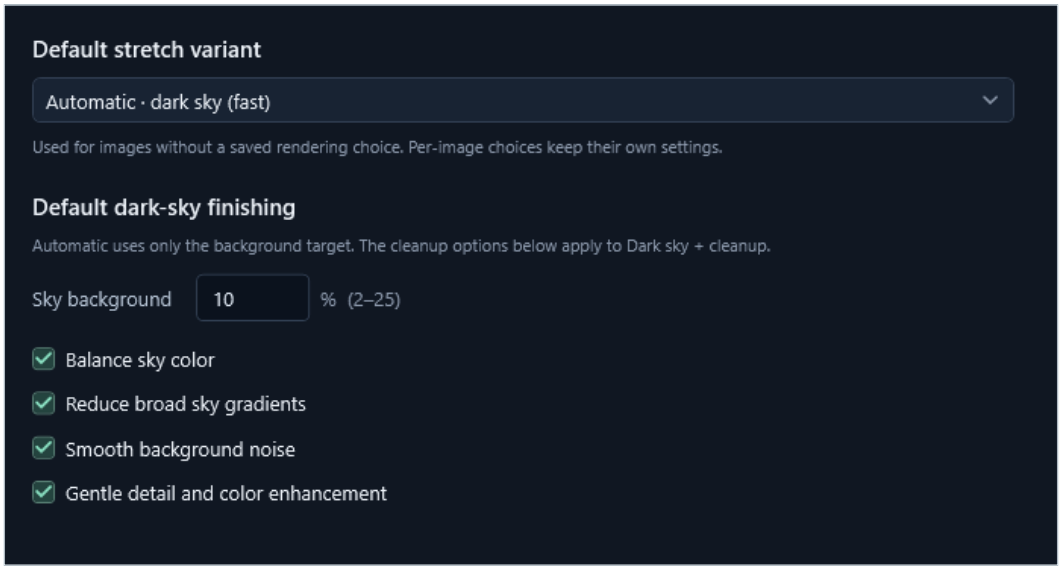
Display choice	When to use it
Automatic · dark sky (fast)	Start here. Uses a stored display function when present; otherwise detects dark linear data and stretches it. It skips cosmetic cleanup.
Linear / already stretched	Use when your image already looks right and should keep its existing exposure.
Library auto stretch (25%)	An alternative, usually brighter automatic display stretch.
Stored display function	Uses the display settings saved inside the image file. Without them, it uses a linear display.
Dark sky + cleanup	Adds the enabled background, noise and detail adjustments. The first processing pass takes longer.

Adjust one image

Expand **Sky background & cleanup** in Selected image. A lower sky-background target makes the background darker. Start near **10%**, then adjust gently. Choose the cleanup options you want and select **Apply finish**. This switches that image to **Dark sky + cleanup**. If a large nebula seems affected, try turning off **Reduce broad sky gradients**.

Choose defaults for future loads

Open **Options > Images**, choose a default stretch and background target, then **Save options**. Automatic uses only the background target, even when the cleanup boxes are checked.



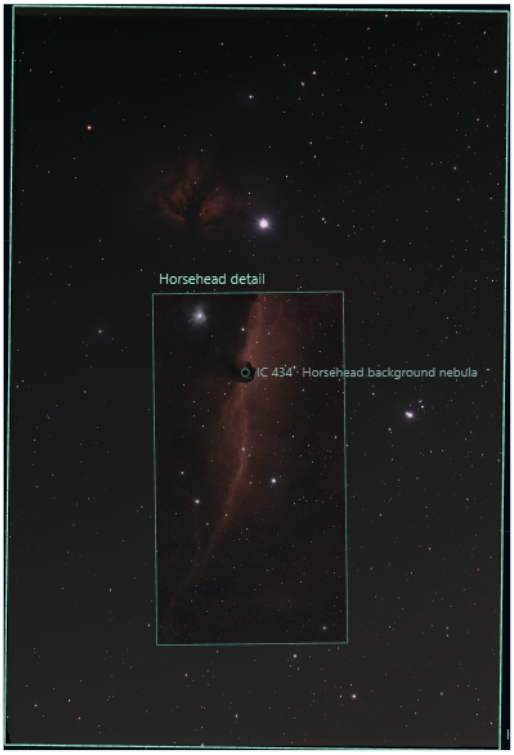
The Images tab with the default Automatic display and a 10% background target. Cleanup boxes apply to Dark sky + cleanup.

Saved per-image choices take precedence over global defaults. Saving Options does not reprocess the images already on screen. **Reset defaults** in Selected image applies your saved image defaults to that image.

05 / Compare overlapping images

SkyBrowser can show a wide photograph and a closer view of the same region together. Matching uses the images' positions and sky coverage, so their filenames do not need to match.

1. Add both photographs. For example, load the wider Horsehead FITS image and the narrower IC434 XISF image from the project's examples folder, if available.
2. Turn on **Blend overlaps** above the sky.
3. Frame either image with **Go to image**. The photograph covering more sky forms the background. A smaller overlapping image becomes a translucent inset.
4. Move the pointer over the inset to reveal it at full opacity. Move away to restore the blend.



Pointer away: the smaller field is blended into the wider background.



Pointer over the inset: its own image is shown at full opacity.

Opacity and ordering

The selected image's **Opacity** slider controls how strongly it appears. With blending enabled, an inset normally uses 60% of that chosen opacity. Hovering temporarily makes it fully opaque; it does not change the saved slider value.

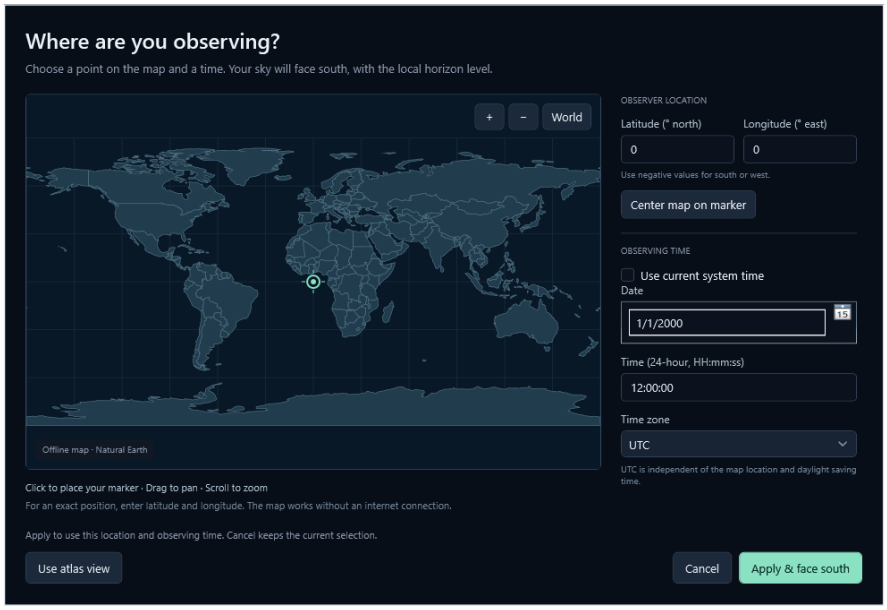
Turn **Blend overlaps** off to use collection order instead. Later entries are drawn above earlier ones; use the arrow buttons to change that order. Hiding a larger background can make its remaining inset opaque when no larger overlapping background remains.

Larger means more sky, not more pixels. A high-resolution close-up can still be the smaller inset. Blending and hovering change transparency only; they do not stretch the image again.

06 / Set your location and time

This is optional. Without a location, SkyBrowser works as a celestial atlas. With a location, it rotates the view to your local horizon and starts facing south.

1. Open **Options > Observer** and choose **Choose on map...**.
2. Click your location on the map. Drag to pan and scroll to zoom. For precision, enter latitude and longitude; negative numbers mean south or west.
3. Leave **Use current system time** checked for a live sky, or uncheck it to choose a fixed date and time.
4. Choose **Apply & face south** to return to Options, then **Save options** to apply and remember the choice.



Demonstration only: latitude 0, longitude 0 and a fixed example time in UTC.

Move through time

With a location selected, **Scene time** appears below the sky. Drag its slider backward or forward up to 12 hours. This pauses the live clock; **Live now** returns to the current system time. **-1 day** / **+1 day** move 24 hours and recenter the slider. With the slider focused, arrow keys move one minute; Page Up/Down move one hour.



The scene timeline, using a neutral example clock shown in UTC.

The readout uses your computer's time zone and UTC offset, not a zone inferred from the map. Options also accepts fixed UTC times. Moving time keeps your direction and zoom and does not reprocess images. The current scene time does not change your saved startup options.

The view starts south, 20 degrees up; **Overview** returns there. **ZENITH** marks straight up (90 degrees). **Polaris** is roughly your latitude above north, reaching overhead only near the North Pole. Use **Clear location > Save options** to return to the atlas and hide the timeline.

> **A simple sky guide:** the map works offline. The horizon does not include mountains, atmospheric effects, daylight or weather. Supported observing dates run from 1900 to 2100.

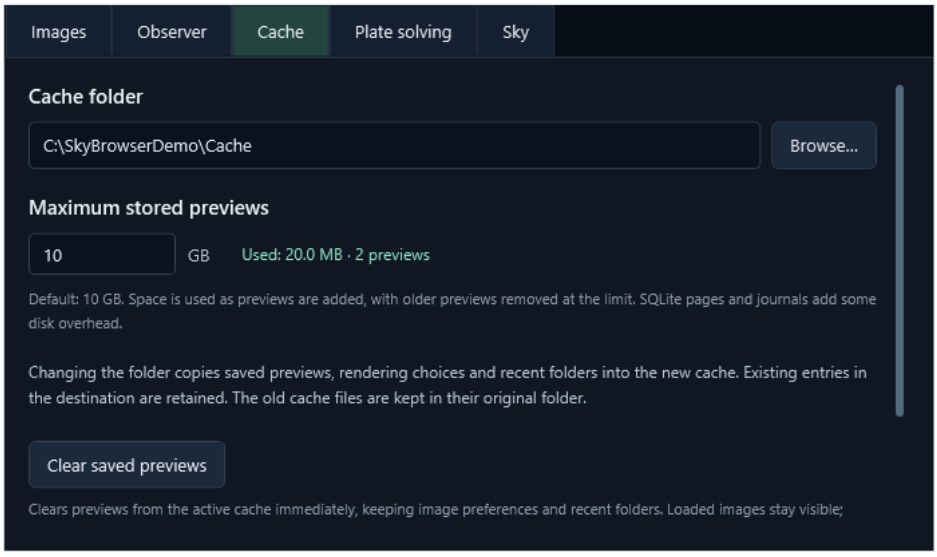
07 / Options and a faster library

Choose **Options** in the top bar. Use **Save options** to apply changes; **Cancel** discards staged changes. The tabs cover:

Tab	What it controls
Images	Default display stretch, sky background and cleanup choices.
Observer	Your optional startup location and live or fixed observing time.
Cache	Where processed previews are stored and how much space they can use.
Plate solving	ASTAP executable/database, time limit and image-height hint.
Sky	Startup grid, constellations, labels, outlines and overlap blending.

Let the cache do the repeated work

The first load may be slow, especially for large images or cleanup. SkyBrowser saves processed previews locally and reuses them on later visits, including after closing and restarting the program. New or changed source files and new display settings can require processing again.



The Cache tab shows the 10 GB default limit and the space already used. Your folder path and usage will differ.

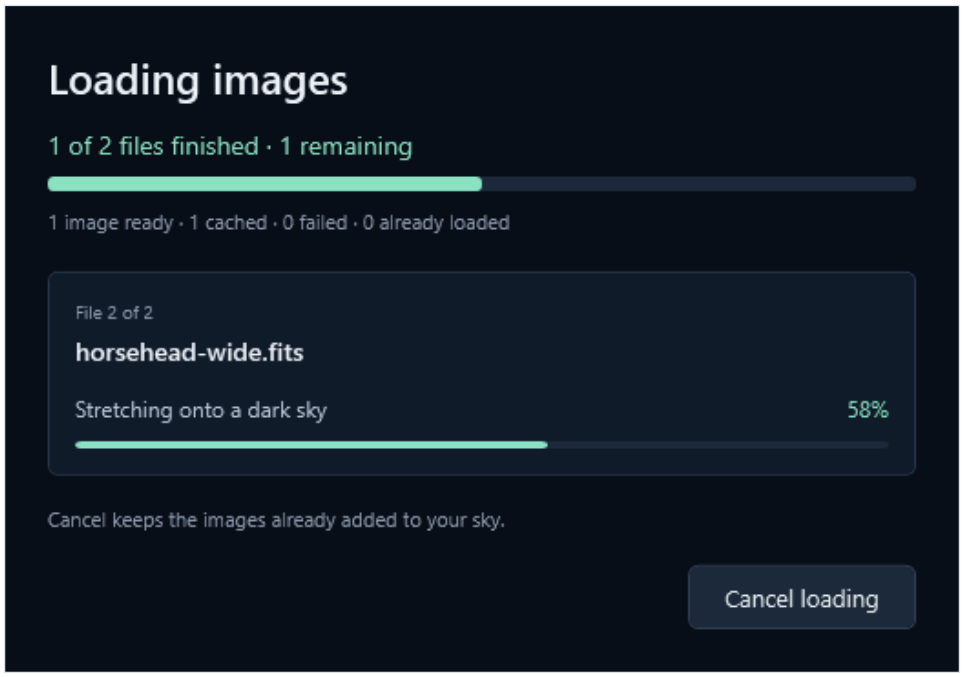
The default limit is **10 GB**. This is a maximum for stored preview data, not space reserved immediately. Older previews are removed as the limit is reached; database overhead can use some extra disk space. Choose a fast local drive with enough free space for a responsive experience.

Changing the cache folder copies cached content when you save. Old cache files remain. **Clear saved previews** clears previews immediately, even if Options is cancelled. It keeps loaded images, image choices, archive metadata and ASTAP solutions; later loads may need processing again.

Where settings live: Options are saved for your Windows user in **%LOCALAPPDATA%\SkyBrowser\settings.json**, separately from the preview database. The Cache tab also shows the options-file path. A cache is disposable; it is not a backup of your images.

08 / Loading and cancellation

Longer loads open a progress window after a short delay. Fast cached visits may finish before it appears.



A loading example: the overall bar counts files; the lower bar shows progress within the current processing step.

The window shows the current filename, completed and remaining entries, the processing step and counts of ready, cached, failed or already-loaded images. Folder imports count files; one file can contain multiple images. A pulsing bar means the current step has no percentage to report.

Choose **Cancel loading**, press Escape in that window, or close it to stop. Some decoding and cleanup steps must finish before cancellation completes. The display shows **Stopping…** while it waits.

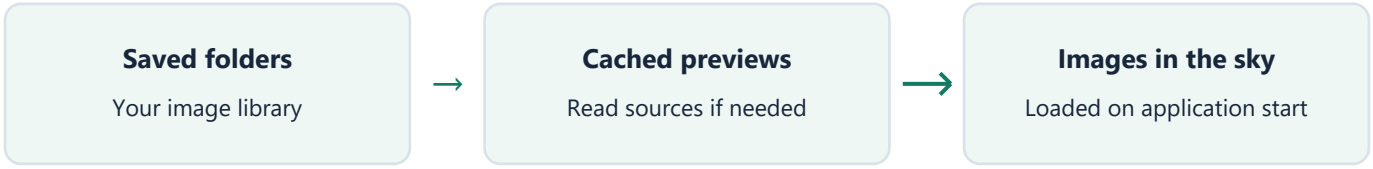
- During an import, images already added stay available; remaining files are not started.
- During a display change, cancellation keeps the previous preview.

If a file fails, other files can still load. Read the message in the bottom status bar and choose **Details** there for more information. The status-bar Details button is an operation log; the small circled i in an image entry opens that image's metadata.

If loading seems slow: let the first visit finish so its previews can be saved. Keep Automatic as your default for quicker processing, and choose cleanup only for images that benefit from it.

09 / Return to your saved folders

SkyBrowser opens the images in your saved library folders automatically when the application starts. You do not need to open a separate file to restore the collection.



Set up your folders once

1. Open **Image library**, add your image folders and choose **Load folders**.
2. Let their images finish loading. Previews and per-image display choices are kept locally.
3. Close the application when finished. Next time, your saved folders load automatically, including subfolders.

What happens on startup

Valid cached previews load first, with their metadata and sky placement, without decoding or stretching the scientific pixels again. A new image, missing preview, changed source or changed XISF sidecar is loaded from its source file and cached for later starts. Newly added files in saved folders are discovered too.

Keep your originals available. Cached previews are validated against the original files and their sidecars. The cache speeds up loading; it is not an image backup. Unavailable folders stay saved and are reported in **Details**.

Stored here	What returns later
Image library	Your folder list and source-bound ASTAP solutions.
Preview cache	Processed previews and per-image display/cleanup choices.
Options	Startup display defaults, sky toggles and observer settings.

Visibility, opacity, hand-entered placement, added target markers and camera adjustments apply to the current sky. Set startup defaults in **Options**. For a portable plate solution, use **Save solved copy...** after blind solving.

Change what opens next time

Use **Image library** → **Remove selected** or **Remove all**, then **Load folders** to confirm. **Cancel** keeps the previous folder list. Individual files opened with **+ Add images** do not add their parent folders to the saved list.

10 / When an image needs placement

Automatic placement requires a stored **astrometric solution**, often called a **plate solve**. It tells the program which sky direction belongs to each part of an image. A target name or acquisition coordinates alone are not enough to determine its width, rotation and possible reflection.

An image marked **Needs placement** has no usable position on the sky. Install ASTAP and a star database, configure **Options** → **Plate solving**, then select the image and choose **Blind solve & place...**. No approximate coordinates are required. The solution is saved locally without rewriting the original. See page 14 for setup and cancellation.

Optional: place it by hand

If you already know the necessary values, select the image and expand **Manual sky placement**. Enter:

Input	Meaning
Center RA	Right ascension of the image center, in decimal degrees from 0 to 360.
Center declination	Degrees north or south of the celestial equator, from -90 to +90.
Width	The full angular width of the image in degrees, not its pixel count.
Rotation	Zero means north up and east left. Positive values turn the image counterclockwise on the sky.
Mirror horizontally	Reverses the image horizontally when needed to match its orientation.

Choose **Apply placement**, then **Go to image**. Inspect alignment against known objects and nearby images. **Use file solution** removes the manual override and returns to the loaded solution, including a retained ASTAP result. Hand-entered placement applies to the current sky.

Manual placement is approximate. A plausible position is not proof of an accurate alignment. Wide or distorted fields especially benefit from a proper plate solve. Moving or stretching a preview does not repair the original file's astrometric metadata.

If a solved image still cannot be placed

Read its warning in Selected image or status-bar **Details**. Supported FITS solutions include SIP/TPV distortion and common zenithal projections; legacy linear PixInsight metadata is recognized too. Other frames, lookup distortions and opaque legacy spline models still need a new plate solution or manual placement. Use the circled i to inspect retained metadata.

11 / Common questions

What you notice	What to try
An image is listed but not visible	Check its visibility checkbox and opacity. Clear the collection search, select it and choose Go to image. If it says Needs placement, follow page 11.
A photograph is too bright	Try Automatic or Linear / already stretched if it was previously processed. For cleanup, lower the sky-background target and select Apply finish.
A photograph is almost black	It may be linear data. Try Automatic or Dark sky + cleanup, and allow processing to finish.
A nebula changes too much with cleanup	Try disabling Reduce broad sky gradients or other cleanup options. Automatic provides a stretch without those cosmetic steps.
Fewer entries appear in the collection	Look for an active search. Clear it with the x button. Filtering the list does not hide photographs from the sky.
A new default does not change a loaded image	Options do not reprocess the current collection. Use Reset defaults in Selected image. Remembered per-image choices can override global defaults.
A file fails to open	Read status-bar Details. Check that it still exists, its format is supported, and any distributed XISF data files are present.
The corner arrow cannot open a source	Check that the original file exists and Windows has a default application for FITS/XISF. Double-click the photograph for internal inspection.
The local sky time seems wrong	Check your computer clock and time zone. For a fixed time, confirm Computer local time versus UTC. The map marker does not choose a time zone.
Loading is slow again after an update	Updated processing, changed sources, a new stretch variant or an evicted preview may need another processing pass. Later cached visits should be quicker.

Know which Details to open

The **circled i beside an image** shows that image's dimensions, source-file facts and metadata. The **Details button in the bottom status bar** shows loading messages and errors. Start with the status-bar log when something fails.

Settings will not save? Check that your Windows user can write to the location shown in Options. For a cache-folder error, choose an accessible local folder. A failed save keeps the options or setup window open with an explanation.

12 / Keep this page handy

Everyday controls

Control	Result
Left-drag / mouse wheel over sky	Look around / zoom.
Arrow keys / + / -	Navigate / zoom while the sky has focus.
F / Home	Frame the selection / return to the overview while the sky has focus.
Ctrl+F	Focus collection search.
Escape in collection search	Clear the search.
Escape in the loading window	Request cancellation.
Double-click a collection entry	Frame that image.
Double-click a photograph in the sky	Inspect full resolution internally.
Corner arrow on a hovered or selected sky image	Open its original in the Windows default application.
Drag the collection divider	Resize the collection and Selected image panels.

A few useful words

- **FITS / XISF:** scientific image-file formats that can store pixels and information about the observation.
- **Metadata:** information saved with an image, such as target, filter, telescope, exposure and coordinates.
- **Stretch:** a display adjustment that makes faint detail visible. It does not change your source image here.
- **RA / Dec:** right ascension and declination, the sky's position coordinates. SkyBrowser's input boxes use decimal degrees.
- **Field of view:** how much sky you can see. A smaller angle gives a closer view.
- **Plate solve / astrometric solution:** information connecting image positions to sky coordinates.
- **Cache:** saved processed previews used to speed up future visits, including after restarting.

A simple routine

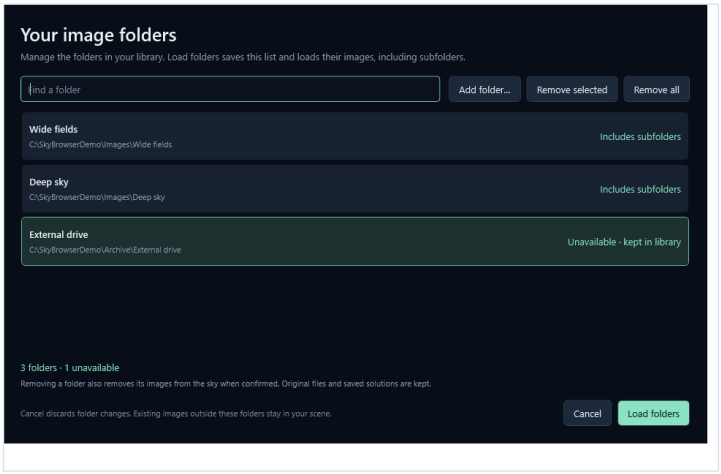
Keep **Automatic** as your default. Browse one folder, select an image and choose **Go to image**. Look around and try the finder; your saved image folders reopen next time. Add a location or cleanup only when you need it.

This guide describes the current Windows desktop application. Screenshots use anonymous demonstration paths, metadata and observing settings. Filenames, values and layout can differ on your computer. Sky-catalogue and map attributions are supplied with the application in **THIRD-PARTY-NOTICES.md**.

13 / Image library and ASTAP

Keep an archive, choose a scene

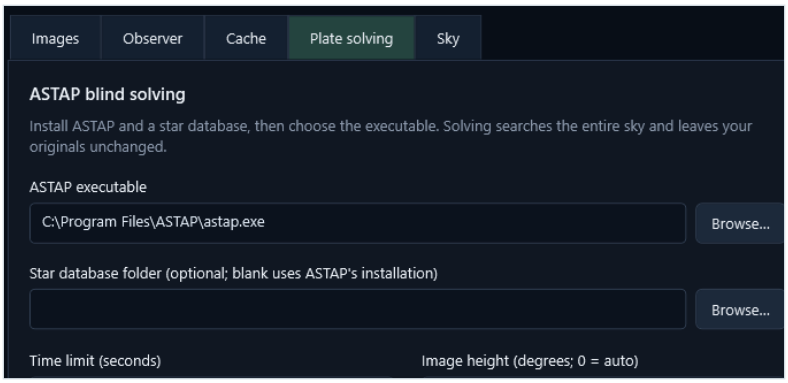
Add folders in **Image library**, then confirm with **Load folders**. Their images, including subfolders, reopen automatically on the next application start.



The library lists image folders with controls to add, remove and confirm them.

Use **Remove selected** (Ctrl/Shift for several) or **Remove all** to edit the list. **Cancel** discards edits. **Load folders** saves all listed folders, even those hidden by the folder filter. Confirmed removals take their images out of the sky; files and saved solutions stay safe. Unavailable folders remain listed for later use.

Solve an image without knowing its coordinates



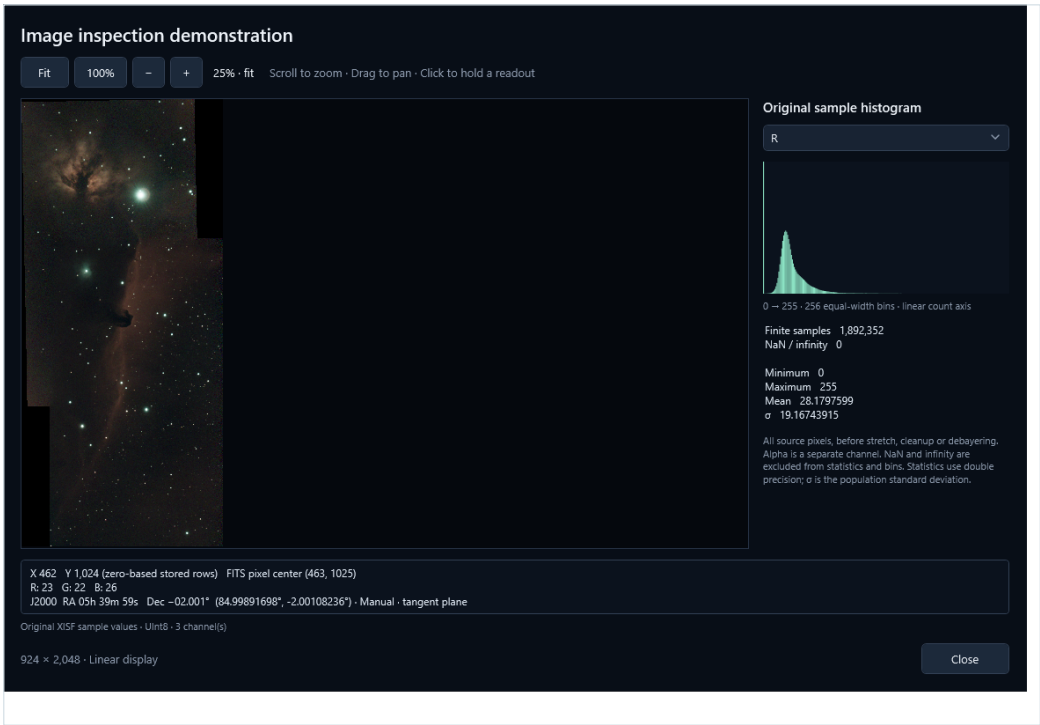
ASTAP executable, optional star database, time limit and image-height hint in Options.

1. Install ASTAP and a star database. In **Options** → **Plate solving**, browse to **astap.exe** or **astap_cli.exe**. The optional database folder is useful for databases installed elsewhere.
2. Keep image height at **0** for automatic scale search, or enter its height in degrees for an unusually wide/narrow field. Save options.
3. Select an image and choose **Blind solve & place…**. ASTAP searches the whole sky. **Cancel loading** or the time limit stops it. The inspector shows fitted field size and pixel scale.
4. Choose **Save solved copy…** and select FITS or XISF to embed the solution in a new file with the original scientific pixels. The source stays unchanged. FITS may omit XISF-only metadata.

Local solutions survive clearing previews; changed source files invalidate them. Saved solved copies carry their astrometry between computers without requiring the local library.

14 / Inspect source pixels

Double-click a photograph in the sky, or choose **Inspect full resolution…** under Selected image or in Image details. This opens the scientific image in its own window with your selected display stretch. The sky keeps its smaller preview.



Full-resolution inspection with a channel histogram and a held pixel readout. This anonymous demonstration uses a separate image made from display pixels.

Zoom, pan and read a pixel

- **100%** maps one source pixel to one screen pixel, including Windows display scaling. **Fit** shows the frame at up to 100%.
- Scroll or use **+ / -** to zoom; drag the image to pan. The sky camera stays where it was.
- Move over the image for zero-based X/Y, one-based FITS pixel centers and original sample values. Click to hold or release the readout for copying.
- J2000 sky coordinates appear when the file, saved ASTAP result or manual placement provides a supported solution. Manual coordinates remain approximate.

Understand the measurements

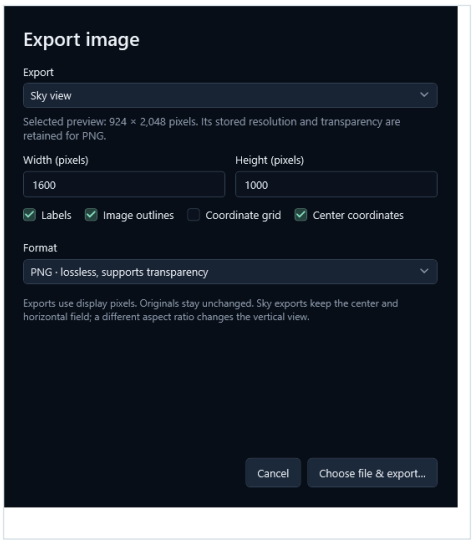
Select a channel for its histogram, minimum, maximum, mean and population standard deviation. All source pixels contribute before stretch, cleanup or debayering. FITS values include BSCALE/BZERO scaling; undefined BLANK samples become NaN. Bayer readouts retain the original mosaic and alpha has its own channel.

The 256 histogram bins span the finite minimum/maximum with a linear count axis. NaN and infinity are counted separately and excluded from the measurements. Aggregate statistics use double precision, so very large integers can round; individual UInt64 sample readouts retain every bit.

Keep the original available. Full-resolution loading needs memory and can take time. Close the window to cancel; a native decoder or filter may finish first. Missing or changing source files produce an explanation without changing your atlas or originals.

15 / Export and sharing

Choose **Export...** in the top bar to save a PNG or JPEG without changing your photographs.



Export settings let you choose the sky or a selected preview, dimensions, overlays and format.

Save the sky view

1. Choose **Sky view** and enter the output width and height in pixels. Each side can be 128–8192 pixels, up to 32 million pixels in total.
2. Choose whether to include **Labels**, **Image outlines**, a **Coordinate grid** and **Center coordinates**.
3. Choose PNG or JPEG, then **Choose file & export...**. Pick the destination and save. For JPEG, quality 92 is a useful starting point.

The export keeps the current center and horizontal field. A different aspect ratio changes how much sky fits vertically. It leaves your live scene unchanged and excludes toolbar controls, corner buttons and temporary hover emphasis.



An anonymous exported sky view with labels, grid and a coordinate caption.

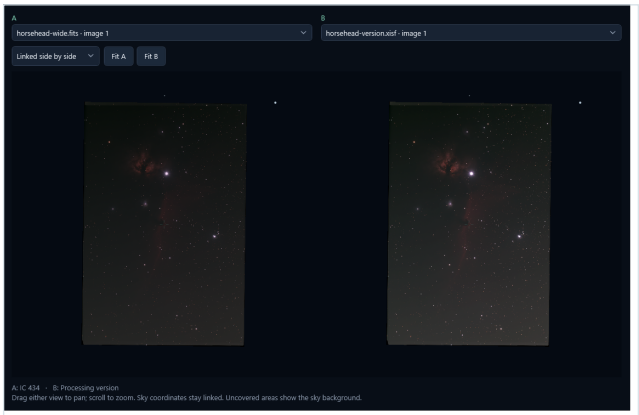
Save one image preview

Select an image first, then choose **Selected image preview** in Export. This saves its current display processing at preview resolution, up to 2048 pixels on the longest side. PNG preserves transparency; JPEG uses a dark background. The original is not decoded again.

Display exports are pictures for sharing. They omit private source metadata and do not carry scientific samples or embedded astrometry. To save original scientific data with a solved position, use **Save solved copy...** instead. Source and XISF sidecar destinations are protected; choose a new filename if sidecars cannot be checked.

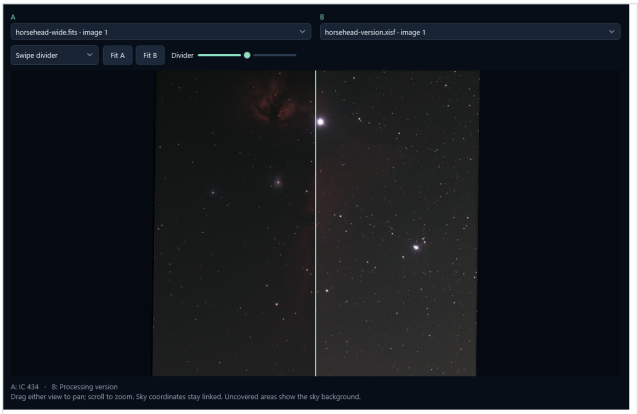
16 / Aligned image comparisons

Select two or more placed collection entries with **Ctrl** or **Shift**, then choose **Compare…** below the collection search. Use the A and B lists to choose a pair from that selection. Unplaced images need a file solution, manual placement or ASTAP result first.



Two processing versions in linked views. The demonstration uses the same image geometry with different display brightness.

Mode	How to use it
Linked side by side	Drag or zoom either pane. Both retain the same sky center and horizontal field.
Aligned blink	Alternate A and B at the selected seconds per frame. Uncheck Blink to pause, or use Space with the sky focused. Show A/B steps between frames.
Swipe divider	Drag the mint divider or use its slider. A appears to the left and B to the right.



The swipe divider reveals two display versions at the same sky coordinates.

Fit A and **Fit B** frame either image. These tools use each image's sky placement, including rotation, mirroring and distortion. Areas outside an image show the sky background. Incorrect manual placements produce incorrect alignment.

Comparison uses the current display previews, up to 2048 pixels on the longest side. It opens independent snapshots at full opacity, including hidden selected images. Closing the window leaves the atlas camera, visibility, opacity and processing unchanged. Reopen it after changing a preview or placement.

17 / Organize your collection

Collection entries now show thumbnails from the loaded previews. Use **Ctrl** or **Shift** to select several entries; **Select all** selects every matching entry, including entries outside the visible scroll area.

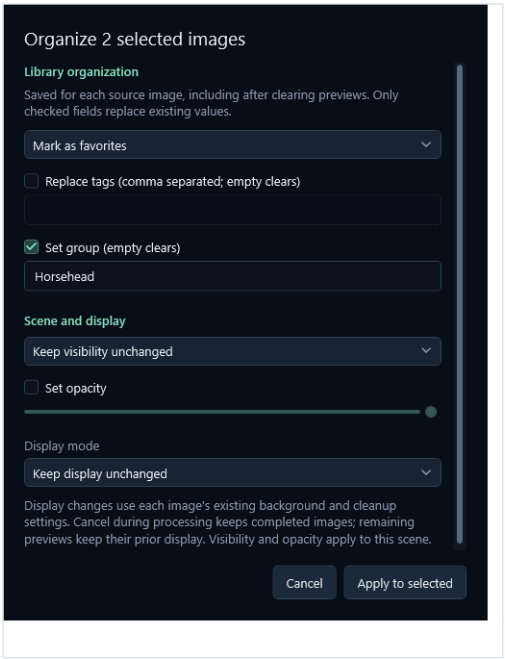
Sort and narrow the list

Expand **Sort & filter**. Sort by name, object, filter, telescope, camera, observation date, favorites or group. **Layer order** returns to the scene's drawing order. Sorting and filtering the list leave the sky layers in place.

Choose a metadata field and enter text it must contain. Observation dates use the source's ISO text, so a year or year-month narrows a date. Tag and group filters match whole names, ignoring case. **Favorites only** combines with all other filters and the existing filename/metadata search. **Clear all filters** restores the complete list.

Apply changes to selected images

Choose **Organize...**. Check only the fields you want to replace. Tags are comma separated; an empty checked field clears tags or group. Each image belongs to one group and may have several tags.



The batch dialog changes only the chosen fields for the selected images.

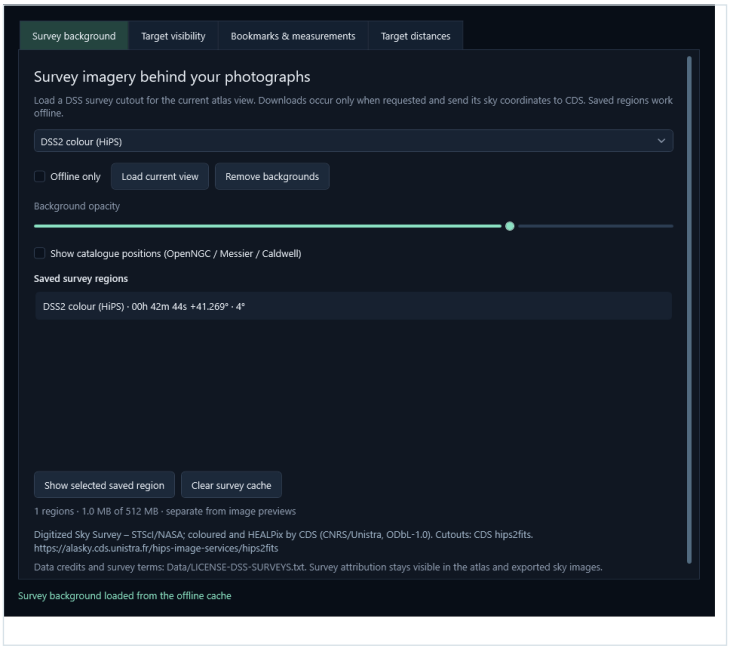
Favorites, tags and groups are saved in the library, per source path and image number. They survive restarts, folder removal and clearing previews. Keep the original paths to retain those associations; moved-file recovery remains future work.

The same dialog can show/hide images, set opacity or change display mode using each image's current background/cleanup settings. Visibility and opacity apply to this scene. Display choices persist. Cancellation keeps saved organization and completed previews; other previews keep their prior display. Failures appear in **Details** and do not stop the remaining batch.

18 / Add a survey background

Open **Sky tools… > Survey background**. DSS images provide context behind your own photographs. They are separate from your collection and never change your scientific sources.

1. Center and zoom the atlas on the region you want.
2. Choose **DSS2 colour (HiPS)** or **DSS2 red (HiPS)**, then **Load current view**. This requests a square, 1024-pixel FITS cutout from CDS using the current sky coordinates. It can take a moment; Cancel retains the previous background.
3. Adjust **Background opacity**. Your photographs remain in front. **Show catalogue positions** adds the offline OpenNGC, Messier and Caldwell overlay; close views show more objects.
4. New cutouts join the displayed regions; newest wins where they overlap. Choose **Remove backgrounds** to return to the star map. Moving the atlas never downloads imagery.



Survey controls and a saved M31 region. Only public demonstration coordinates are used.

Use saved regions offline

The separate survey cache keeps up to **512 MB** of downloaded FITS payloads. Select a region under **Saved survey regions**, then **Show selected saved region** to load it without a connection and frame its position. **Offline only** prevents network access; loading an uncached view explains that it is unavailable.

Clear survey cache removes saved regions immediately, retaining the displayed background and your image previews. Older survey regions are evicted as the limit is reached. A missing or damaged region can be downloaded again when online.

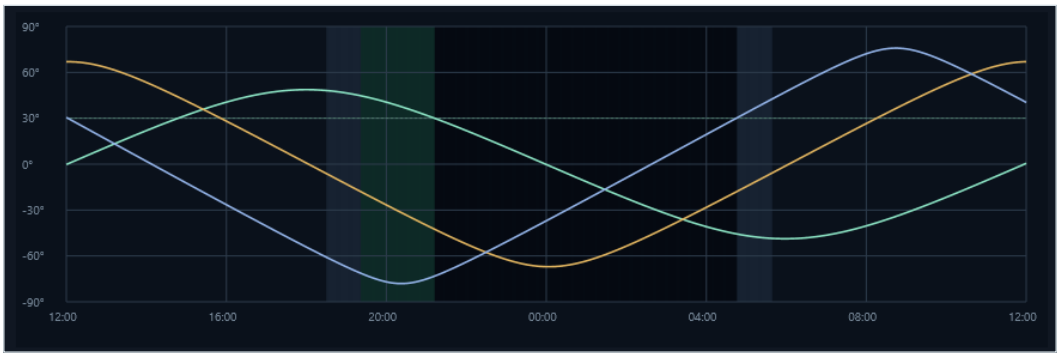
Enable **Options > Sky > Show cached DSS2 images by default** to restore all saved regions behind your photographs at startup. It also applies when you save Options, preserves your view, and never downloads. The option is off initially; damaged regions are skipped.

Retain survey credits when sharing. The atlas and sky exports show DSS/STScI/NASA and CDS attribution even with Labels off. Full provider acknowledgments and ODbL-1.0 references are supplied in **Data/LICENSE-DSS-SURVEYS.txt**. The application license does not relicense downloaded survey data.

19 / Plan target visibility

Open **Sky tools… > Target visibility**. The planner works offline and uses a temporary location without changing your startup options.

1. Choose a target in the atlas and select **Use atlas center**, or enter its J2000 RA and declination in degrees.
2. Enter latitude (north positive) and longitude (east positive), or choose **Use observer location** when one is active.
3. Choose the start date and time in **UTC**. The report covers the next 24 hours, including across computer-local daylight-saving changes.
4. Set a minimum target altitude, maximum Sun altitude and minimum Moon separation. Start with **30°**, **-18°** and **30°**. Optionally ignore Moon separation while the Moon is below the horizon, then **Calculate visibility**.



Target, Sun and Moon altitude curves. Green shading marks imaging windows; darker bands mark twilight and astronomical darkness.

Read the chart and times

The mint curve is your target; gold is the Sun and blue is the Moon. The dashed line is your altitude threshold. Hover over a time for altitude, Moon separation and illumination readouts. Green intervals satisfy all three imaging criteria.

```
2000-01-01 12:00 – 2000-01-02 12:00 UTC
J2000 00h 42m 44s +41.269° · latitude 0°, longitude 0°
Rises and sets in this period. Maximum altitude 48.7°.
Rise (geometric horizon): 01-01 12:00 UTC
Set (geometric horizon): 01-01 23:58 UTC
Rise (geometric horizon): 01-02 11:56 UTC
Upper transit: 01-01 17:59 UTC
Moon at period midpoint: Waning crescent · 19% illuminated · separation 136.0° · altitude -37.0°
Darkness (Sun ≤ -18°):
  01-01 19:21 – 01-02 04:45 UTC (9.39 h)
Imaging windows (altitude, darkness and Moon criteria):
  01-01 19:21 – 01-01 21:12 UTC (1.85 h)
```

A fixed demonstration report for M31, latitude 0 and longitude 0, beginning at noon UTC on 1 January 2000.

The report lists geometric rise/set events, upper transit, maximum altitude, Moon phase/illumination and separation, darkness windows and imaging windows. Circumpolar targets and polar day/night produce explicit results instead of invented crossings.

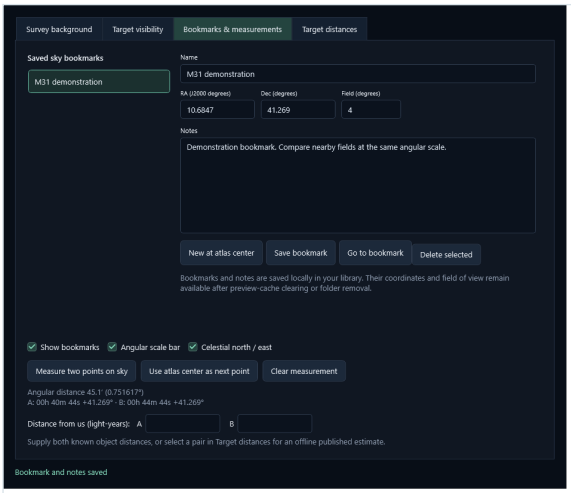
Interpret the limits. Altitudes assume a flat horizon at sea level, without refraction, terrain or weather. Curves and detection of short windows use two-minute samples; detected boundaries are refined. Very brief grazing windows can be missed. A useful geometric window does not guarantee clear skies or suitable local conditions.

20 / Bookmarks and measurements

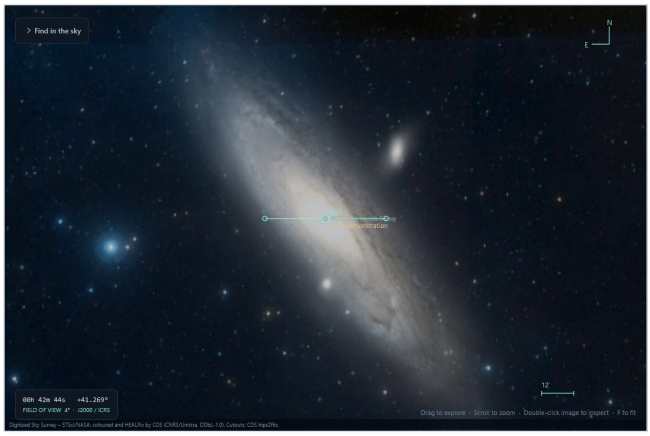
Open **Sky tools… > Bookmarks & measurements** to keep named sky positions and notes, or measure separations on the celestial sphere.

Save a place and its notes

1. Choose **New at atlas center**, enter a name and notes, then **Save bookmark**. You can edit its RA, declination and field of view before saving.
2. Select a saved bookmark to read or update it. **Go to bookmark** restores that position and zoom; **Delete selected** removes it.
3. **Show bookmarks** toggles their sky markers. Labels controls their names. Bookmarks and notes survive restarts, preview-cache clearing and folder removal; source images remain unchanged.



The bookmark editor and a fixed demonstration note.



M31 in the public DSS survey with a measurement, celestial directions and angular scale. Survey credits remain at the bottom.

Measure and orient the sky

Choose **Measure two points on sky**, then click a start and end in the atlas. Moving the pointer previews the arc after the first click. The result gives the shortest great-circle separation in degrees, arcminutes or arcseconds, including across the RA seam and near poles. **Escape** cancels an unfinished measurement; **Clear measurement** removes it.

For keyboard use, navigate the atlas and choose **Use atlas center as next point** twice. Measurements apply to the current scene and are not saved automatically; copy values into a bookmark note if needed.

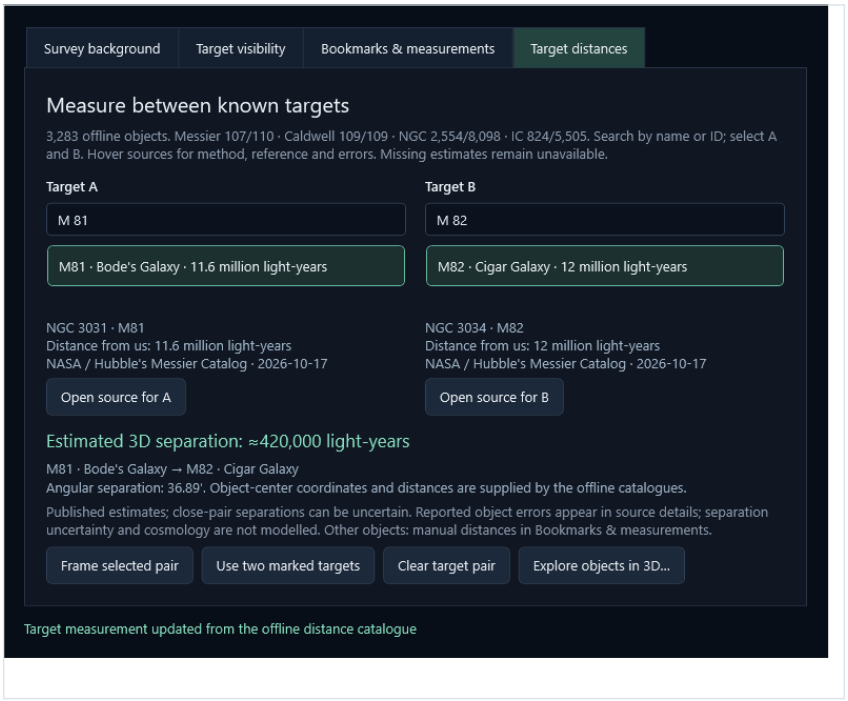
After choosing both points, enter their known **Distance from us (light-years)** under **A** and **B** for a straight-line 3D separation. For prepared object distances, use **Target distances** (page 22). For example, two objects each 1,000 light-years away and 1° apart are about 17.45 light-years apart. Both manual distances must be positive; leaving either blank keeps only the angle. New/cleared measurements reset the inputs. The Euclidean estimate inherits distance uncertainties and does not model cosmological distances.

Angular scale bar measures angles at its displayed position. **Celestial north / east** follows J2000 sky directions through the current projection, including observer-mode rotation. It is separate from the horizon compass and is hidden at an exact celestial pole, where north has no unique direction.

21 / Use published target distances

Open **Sky tools… > Target distances** to measure between known objects without typing distances. The offline list contains **3,283 unique objects** from NASA and SIMBAD/CDS: **107/110 Messier**, **109/109 Caldwell**, **2,554/8,098 NGC** and **824/5,505 IC** entries. Aliases share estimates, so counts overlap. Missing estimates remain unavailable.

1. Search under **Target A** by designation or common name: **M 81**, **NGC 3031**, **Caldwell 102** or **IC 2602**. Select the result.
2. Search under **Target B**, for example **M82** or **Cigar**, and select it. The result and atlas measurement update automatically using the objects' catalogue centres.
3. Choose **Frame selected pair** to see both endpoints. Selection alone preserves your view. If you already added two eligible markers with **Find in the sky**, use **Use two marked targets** instead.



The expanded offline list shows coverage for all four catalogues. Each selected object retains its source; separation is approximate.

Read the estimate and its sources

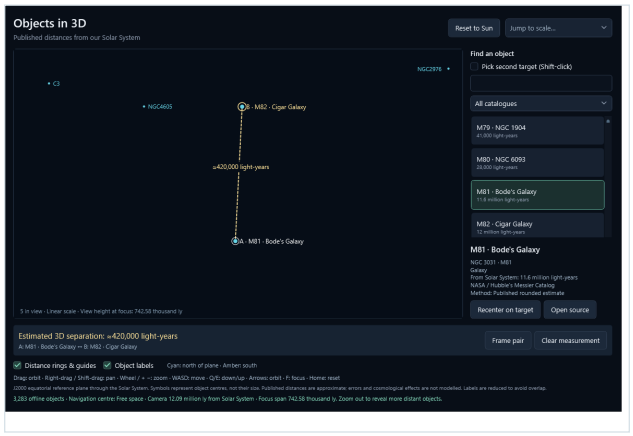
The NASA values for M81 and M82 are **11.6 million** and **12 million light-years** from us, giving an estimated separation of **about 420,000 light-years**. **Open source for A/B** opens the cited page in your browser. Hover over source text for its method, literature reference and reported object errors. Selection and calculation work offline.

> **An estimate, not an exact measurement.** The app marks separation with ≈ and rounds it to two significant figures. Sources differ in method and precision; error bars are shown when supplied, but are not propagated to separation errors. Rounding can strongly affect close pairs such as M81/M82. The calculation uses Euclidean geometry without a cosmological model or inferred redshift distances.

Objects outside this list can still use manual inputs in **Bookmarks & measurements**. Manual edits detach the catalogue source; starting a new measurement or **Clear target pair** clears the selected pair. Ordinary sky clicks never assume an object identity. Measurements and selections apply to the current scene and are not saved automatically.

22 / Explore objects in 3D

Open **Sky tools… > Target distances > Explore objects in 3D…**. Explore 3,283 objects using their sky directions and published distances, starting at our **Solar System**. This works offline.



M81 and M82 connected by their estimated straight-line separation in light-years.

Move through space

Control	Action
Left-drag	Orbit around the current focus
Right-drag or Shift-drag	Pan the focus and camera together
Mouse wheel or + / -	Zoom in or out
W/S, A/D, Q/E	Move forward/back, left/right, down/up
Arrow keys / F	Orbit / focus the selected object
Home or Reset to Sun	Restore the starting view

Click the map for keyboard controls; hold **Shift** to move faster. **Jump to scale…** changes scale around the current centre. Scroll out to reveal more distant objects.

Find an object and read the scale

Click a point or search result to centre zoom and orbit on it. **Recenter on target** returns after panning; **Reset to Sun** restores the starting view. Search filters the list; the catalogue choice filters both list and map. **Open source** opens the citation; hover for errors and references.

Measure a pair: Click A, then **Shift-click B**, or enable **Pick second target** and choose B. The gold line shows their estimated separation in light-years. **Frame pair** fits both points; **Clear measurement** or **Escape** removes B. A normal new selection or reset clears the pair.

Rings measure **light-years from the Solar System** in the **J2000 equatorial plane**. Cyan points are north of it; amber points are south. Guides connect labelled objects to the plane. Toggle guides and labels below the map; labels are reduced to limit overlap.

Scale and limits: Linear positions; symbol sizes. Distance errors and cosmological effects are not modelled. Navigation leaves the atlas unchanged and resets on closing.